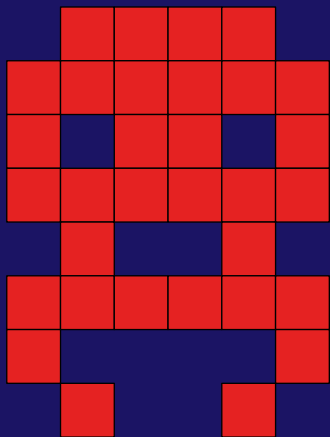
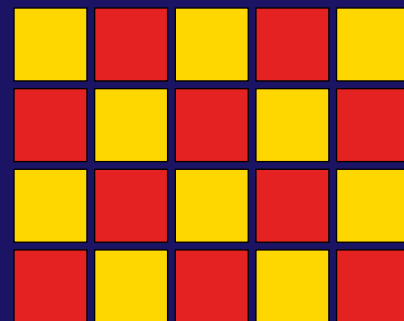




INTO THE PIXEL GRID

像素格世界

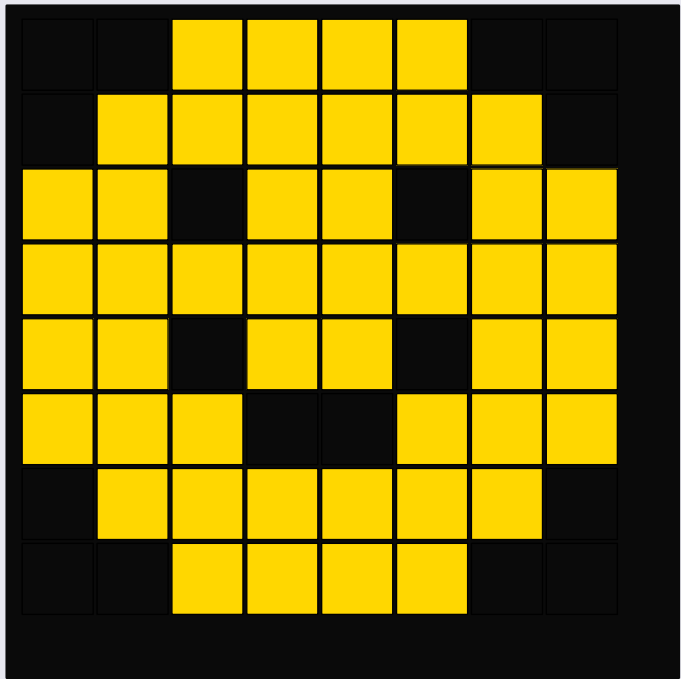
iPad Digital Art · Session 4



ICS Creator Academy · Summer 2026

WHAT IS A PIXEL? 像素是什麼？

PIXEL = Picture + Element 圖片的最小單位！



Zoom in! 放大看！



Pixel = one tiny square 一個小方格

Every digital image is made of millions of tiny squares called pixels. Each square has exactly ONE colour.



Screens are pixel grids 螢幕是像素網格

Your iPad screen has millions of pixels! 2048×1536 pixels on a standard iPad — that's 3,145,728 tiny squares!



8-bit = 8 bits of colour 8位元色彩

Old games could only use 256 colours. Each pixel was described by 8 'bits' of data — that's why it's called 8-bit 八位元!



Pixel art is intentional 像素畫是刻意的

Modern games have millions of colours. Pixel art CHOOSES to use big visible pixels as an artistic style — retro and beautiful!

THE 8-BIT ERA 八位元時代

When video games were born 電玩遊戲誕生的年代

1972



Pong — The First Game

The very first video game! Just two rectangles and a square. Two pixels bouncing — and the world went crazy! 世界上第一個電玩遊戲！

1980



Space Invaders & Pac-Man

Pixel aliens and a yellow dot-eating character took over arcades worldwide. 小精靈跟外星人入侵！Pac-Man made more money than Star Wars that year!

1983



Super Mario Bros.

Nintendo's masterpiece! Mario was designed in pixels because of hardware limits — but became the most famous video game character EVER. 超級瑪利歐！

1989



Game Boy

Nintendo's portable console — only 4 shades of green! Kids worldwide played Tetris and Pokémon on tiny pixel screens. 任天堂Game Boy！

Today
今天



Pixel Art Revival 像素藝術復興

Games like Minecraft, Stardew Valley, and Undertale CHOOSE pixel art. It's not a limit anymore — it's a style! 現在是一種藝術風格！

FAMOUS PIXEL ART CHARACTERS 著名像素藝術角色

Each one started as a tiny grid of coloured squares! 每一個都是從小小的格子開始的！

Mario 瑪利歐

Super Mario Bros. 1983

Mario was originally called 'Jumpman'. His moustache was added because faces were too hard to draw in pixels! 鬍子是因為太難畫臉！

Space Invader 太空侵略者

Space Invaders 1978

Designed by Tomohiro Nishikado in just one week. He based the aliens on H.G. Wells' 'War of the Worlds' 世界大戰！

Pac-Man 小精靈

Pac-Man 1980

Inspired by a pizza with a slice missing! 設計師看著缺了一片的披薩得到靈感 The designer Toru Iwatani got the idea from looking at a pizza.

Link 林克

The Legend of Zelda 1986

Originally just 16×16 pixels. The tiny green elf became one of gaming's most beloved heroes! 只有16×16個像素！

Pikachu 皮卡丘

Pokémon Red/Blue 1996

The original Game Boy Pikachu was just a yellow blob! The cute design we know came later. 原版只是黃色的小圓形！

Steve 史蒂夫

Minecraft 2009–present

Steve's face is just 8×8 pixels — but Minecraft has sold 238 million copies! The best-selling game ever. 全球最暢銷的遊戲！

HOW TO MAKE PIXEL ART 如何製作像素畫

1

Choose your canvas size

選畫布大小

Small = more pixel, more retro feel.
16×16 is great for beginners! 越小越復古
· 16×16 適合初學者!

2

Plan your colours 規劃顏色

Limit yourself to 4-8 colours only! Real 8-bit games had very few colours — constraints make you creative! 限制顏色數量!

3

Start with the outline 從輪廓開始

Draw the basic shape first — just the outline. Fill in colours later. Keep it simple! 先畫輪廓 · 再填色!

4

Fill & shade 填色與陰影

Fill main colours first. Add one darker shade for shadows, one lighter for highlights. That's all you need! 暗色做陰影 · 亮色做高光!

5

No anti-aliasing! 不要模糊邊緣

Keep edges sharp and blocky — that's the pixel art style! Avoid blur filters. Embrace the squares! 保持邊緣清晰方塊感!

6

Zoom in to draw, zoom out to check

Work zoomed IN (×8 or more) to place pixels. Zoom OUT regularly to see the full image. 放大畫 · 縮小看!

YOUR APP TODAY 今天用的 App



8bit 像素畫家

8-bit Pixel Painter

Free App on iPad App Store

Simple · Fun · Perfect for beginners
簡單 · 好玩 · 初學者最適合



Pencil tool 鉛筆

Tap once = one pixel. Drag = draw a line of pixels. Your main tool! 最常用的工具！



Fill tool 填色桶

Tap an area to fill it with colour instantly. Perfect for backgrounds! 一鍵填滿顏色！



Eraser 橡皮擦

Remove pixels one by one. Drag to erase quickly. 一個一個移除像素！



Zoom 縮放

Pinch to zoom in and draw carefully. Zoom out to see the full picture! 捏合縮放！



YOUR TURN! 換你了！

PIXEL ART CHALLENGE 像素畫挑戰

Open 8bit 像素畫家 on your iPad → New Canvas → Start!

1 Choose a subject 選一個主題

A simple object: 🍏 apple · 🏠 house · ★ star · ❤️ heart
OR your favourite game character! 選你喜歡的遊戲角色！

2 Max 8 colours 最多8種顏色

Pick your colour palette first.
Constraints = creativity! 限制讓你更有創意！

3 Start small 從小開始

Use a 16×16 or 32×32 canvas.
Bigger = harder for beginners! 初學者選小一點的畫布！

4 Share your work 分享你的作品

Screenshot and show the class!
Explain: what did you make? Why? 說說你畫了什麼！